

Salesforce CRM Project Documentation

What Next Vision Motors: Shaping the Future of Mobility with Innovation and Excellence

1. Project Overview

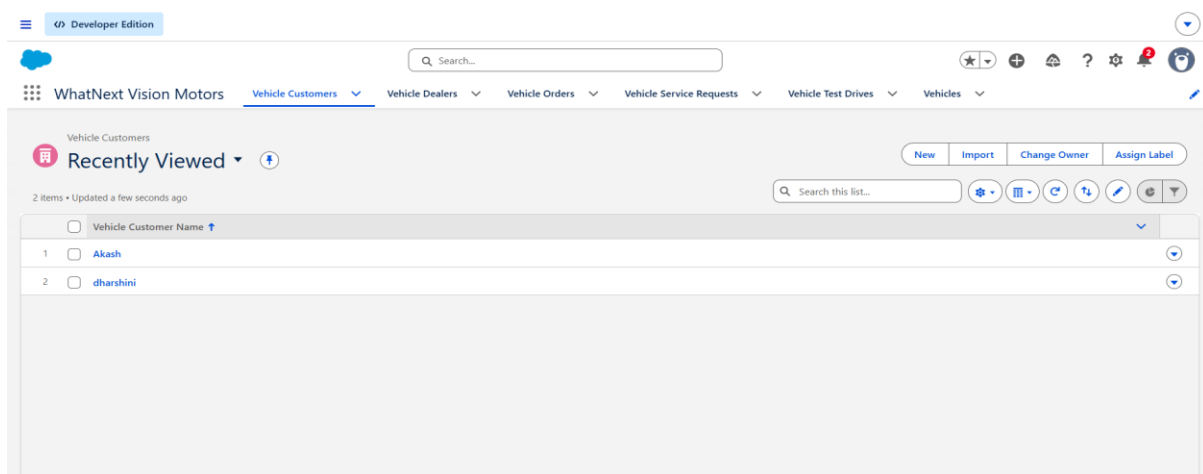
WhatNext Vision Motors is a purpose-built Salesforce CRM application designed to bring together the day-to-day operations of an automotive dealership network under a single platform. Rather than tracking vehicles, customers, dealers, orders, test drives, and service requests through disconnected spreadsheets or systems, the application gives every stakeholder one consistent place to view and update this information.

At the core of the solution is the vehicle-ordering journey. Before any order is confirmed, the system checks that the requested vehicle is actually in stock, and it can automatically match the order to the right dealer based on rules configured in the org. To cut down on manual follow-up, the application also automates recurring activities — reminding customers ahead of scheduled test drives and pushing forward orders that were left pending. On top of this, the Lightning interface gives users a clean, organized way to move between records, and built-in reports and dashboards give managers a quick read on how the business is performing.

2. Objectives

The project was undertaken with the following goals in mind, each of which ties directly back to a measurable improvement in dealership operations:

- Centralize automotive information — bring vehicle, customer, dealer, order, test-drive, and service data into one Salesforce environment so every team works from a single source of truth.
- Improve order processing — give staff a consistent, structured way to create and manage vehicle orders while factoring in real-time vehicle availability.
- Reduce manual activity — apply Salesforce Flow and Apex to automate repetitive operations, freeing staff to focus on customer-facing work.
- Maintain data accuracy — use validation rules and programmatic checks to catch incorrect or incomplete transactions before they are saved.
- Simplify user interaction — deliver a Lightning application with organized navigation, well-structured page layouts, and record pages that surface the most relevant information.



Phase 1: Requirement Analysis and Planning

The opening phase of the project focused on converting the automotive business scenario into a set of concrete, implementable Salesforce requirements — covering requirement analysis, project scope, and the data and security models the rest of the build would rely on.

Business Requirements

Requirements were identified by studying the full range of activities carried out by an automotive dealership, both before and after a vehicle sale. The system needed to hold accurate vehicle and customer information while supporting the transactional processes built around them. The key requirements identified were:

1. Store detailed vehicle information.
2. Maintain accurate customer records.
3. Maintain dealer information across the network.
4. Record every vehicle order placed.
5. Associate each order with its corresponding customer, vehicle, and dealer.

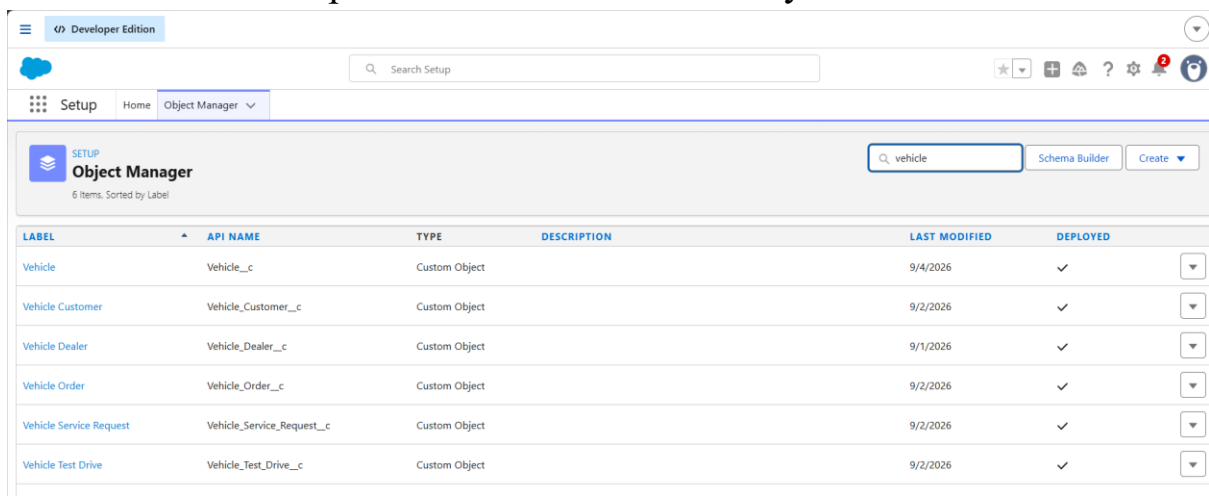
Project Scope

- Vehicle Management — maintains core vehicle information including model details, pricing, stock availability, and related data.
- Dealer Management — stores and manages dealership information to support sales and dealer-related activity.
- Customer Management — maintains customer details required for sales, test drives, and service interactions.
- Order and Test Drive Management — records purchase orders and manages test-drive bookings by connecting customers to their selected vehicles.

- Service Request Management — records customer service needs and links them to the appropriate customer and vehicle records.

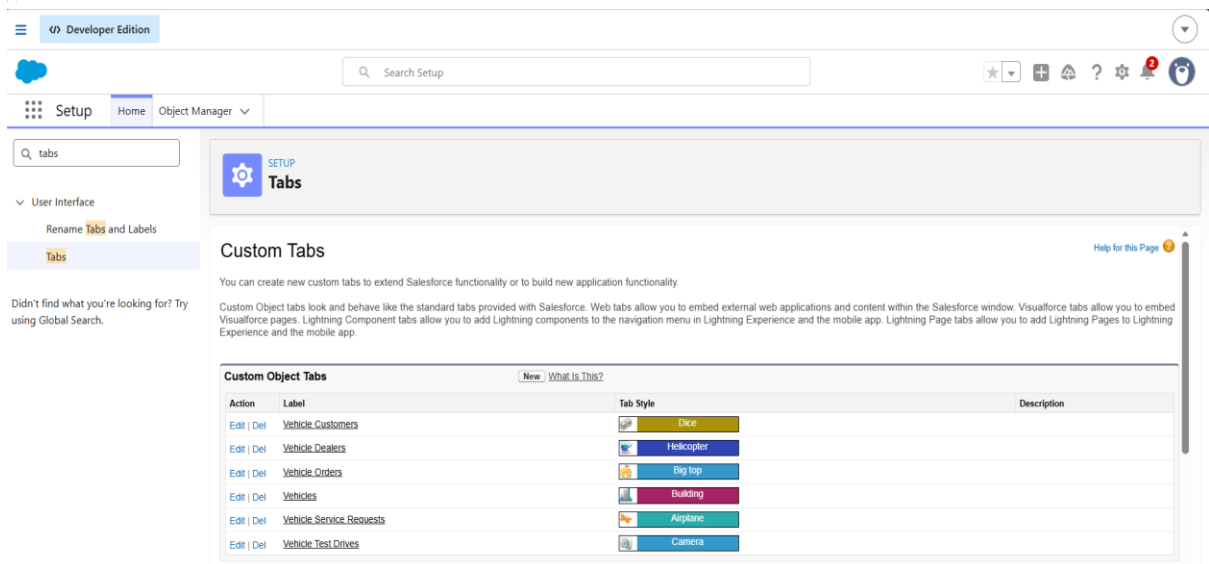
Data Model

The data model captures how the major entities in the automotive process relate to one another. Vehicle records connect to both order and test-drive records; customer information links to the orders and test drives associated with that customer; and dealer records tie into the relevant vehicle transactions. Structuring the data this way removes the need to duplicate information and keeps related records within easy reach of one another.



The screenshot shows the Salesforce Object Manager interface. At the top, there's a search bar with "vehicle" entered. Below the search bar, a table lists six custom objects. Each row includes a label, API name, type, description, last modified date, and deployment status.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Vehicle	Vehicle__c	Custom Object		9/4/2026	✓
Vehicle Customer	Vehicle_Customer__c	Custom Object		9/2/2026	✓
Vehicle Dealer	Vehicle_Dealer__c	Custom Object		9/1/2026	✓
Vehicle Order	Vehicle_Order__c	Custom Object		9/2/2026	✓
Vehicle Service Request	Vehicle_Service_Request__c	Custom Object		9/2/2026	✓
Vehicle Test Drive	Vehicle_Test_Drive__c	Custom Object		9/2/2026	✓



The screenshot shows the Salesforce Custom Tabs page. It includes a search bar with "tabs" entered. Below the search bar, there's a section titled "Custom Tabs" with a description of how to create new custom tabs. Below this, a table lists custom object tabs with columns for Action, Label, Tab Style, and Description.

Action	Label	Tab Style	Description
Edit Del	Vehicle Customers	Dark	
Edit Del	Vehicle Dealers	Helicopter	
Edit Del	Vehicle Orders	Big top	
Edit Del	Vehicles	Building	
Edit Del	Vehicle Service Requests	Airplane	
Edit Del	Vehicle Test Drives	Camera	

Security Model

Security considerations were factored into the design from the start, using Salesforce's layered access-control model:

- Profiles
- Roles
- Role hierarchy
- Permission sets
- Object-level permissions

Phase 2: Salesforce Development — Backend & Configurations

This phase covered the implementation of the underlying data structures and backend functionality that power the application — objects, fields, validation, automation, and Apex.

Setup Environment & DevOps Workflow

Development was carried out entirely within Salesforce Lightning Experience, which provided the environment for creating custom objects, configuring fields and relationships, building automation, developing Apex components, generating reports, and managing security.

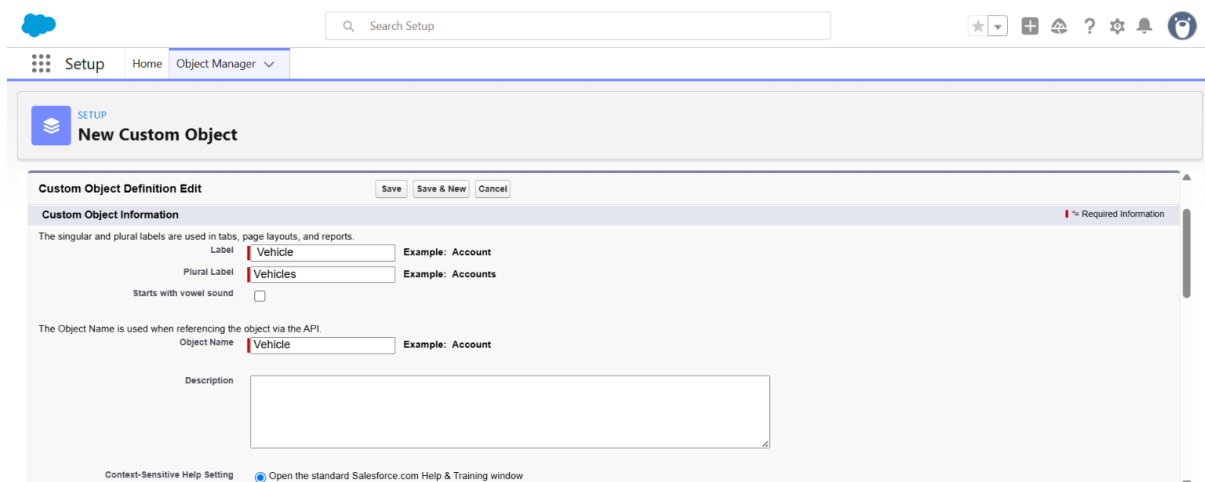
Customization of Objects, Fields, Validation Rules and Automation

1. Custom Objects

Six custom objects represent the core business entities of the WhatNext Vision Motors system:

- **Vehicle** — stores vehicle details such as model, price, stock quantity, availability, and dealer association.
- **Dealer** — maintains dealership details including name, location, city, and contact information.

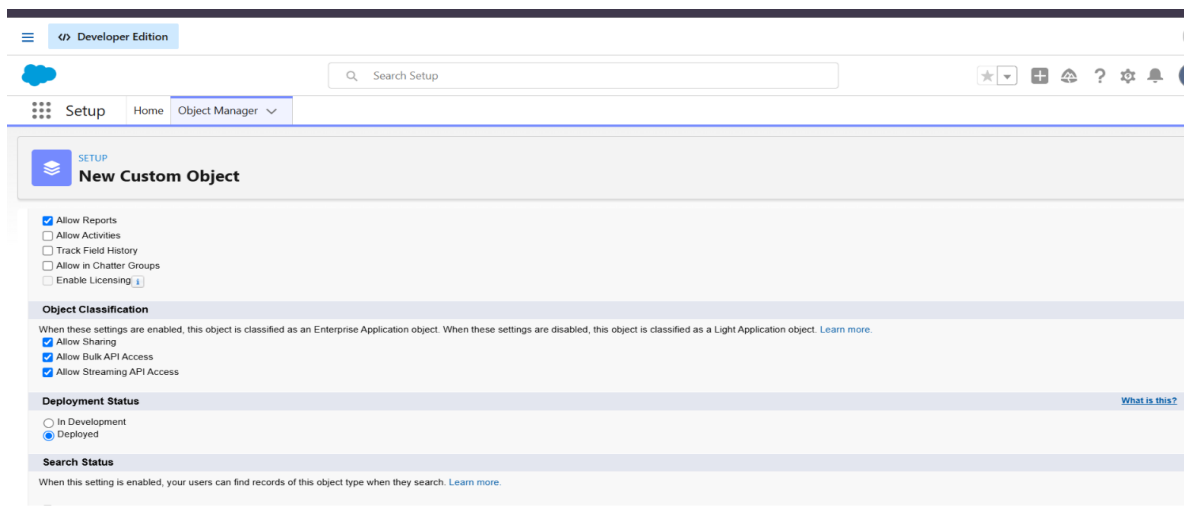
- **Customer** — stores customer information required for vehicle purchases, test drives, and service activity.
- **Vehicle Order** — records customer vehicle purchases and tracks the selected vehicle, customer, dealer, and order status.
- **Vehicle Test Drive** — manages test-drive bookings, tracking customer, vehicle, appointment date, and status.



The screenshot shows the 'New Custom Object' page in Salesforce Setup. The page is titled 'New Custom Object' and has a 'Custom Object Definition Edit' header. Below the header, there are three tabs: 'Save', 'Save & New', and 'Cancel'. The main section is 'Custom Object Information'. It contains the following fields:

- Label:** Vehicle (Example: Account)
- Plural Label:** Vehicles (Example: Accounts)
- Starts with vowel sound:** ☐
- Object Name:** Vehicle (Example: Account)
- Description:** (Empty text area)

At the bottom, there is a 'Context-Sensitive Help Setting' section with a link to 'Open the standard Salesforce.com Help & Training window'.



The screenshot shows the 'New Custom Object' page in Salesforce Setup, displaying configuration options. The page is titled 'New Custom Object' and has a 'Developer Edition' header. Below the header, there are three tabs: 'Setup', 'Home', and 'Object Manager'. The main section is 'New Custom Object'. It contains the following sections:

- Allow Reports:**
 - ☒ Allow Reports
 - ☐ Allow Activities
 - ☐ Track Field History
 - ☐ Allow in Chatter Groups
 - ☐ Enable Licensing [1](#)
- Object Classification:**

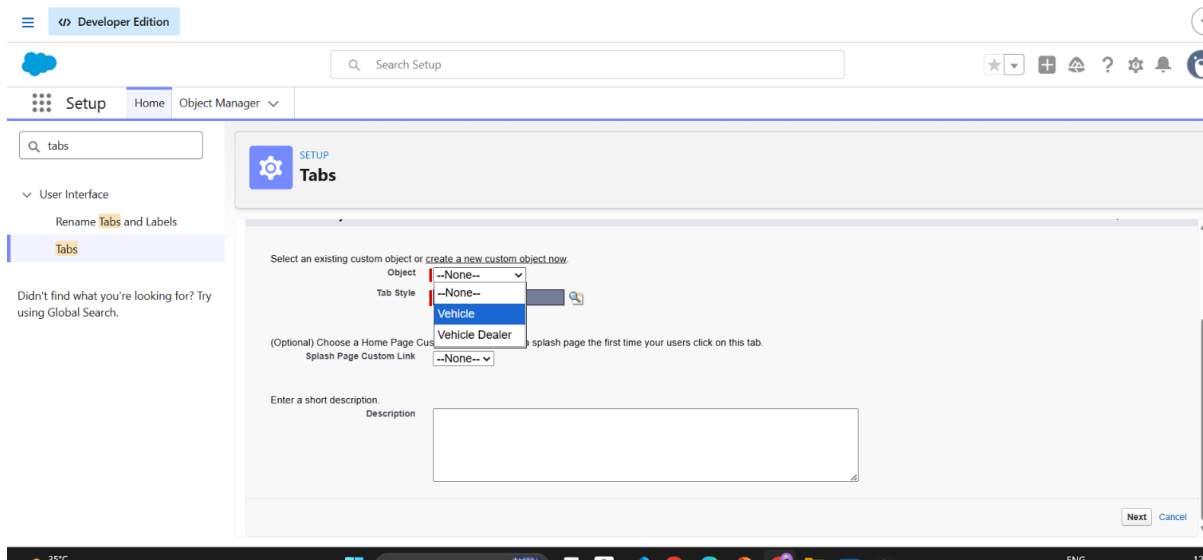
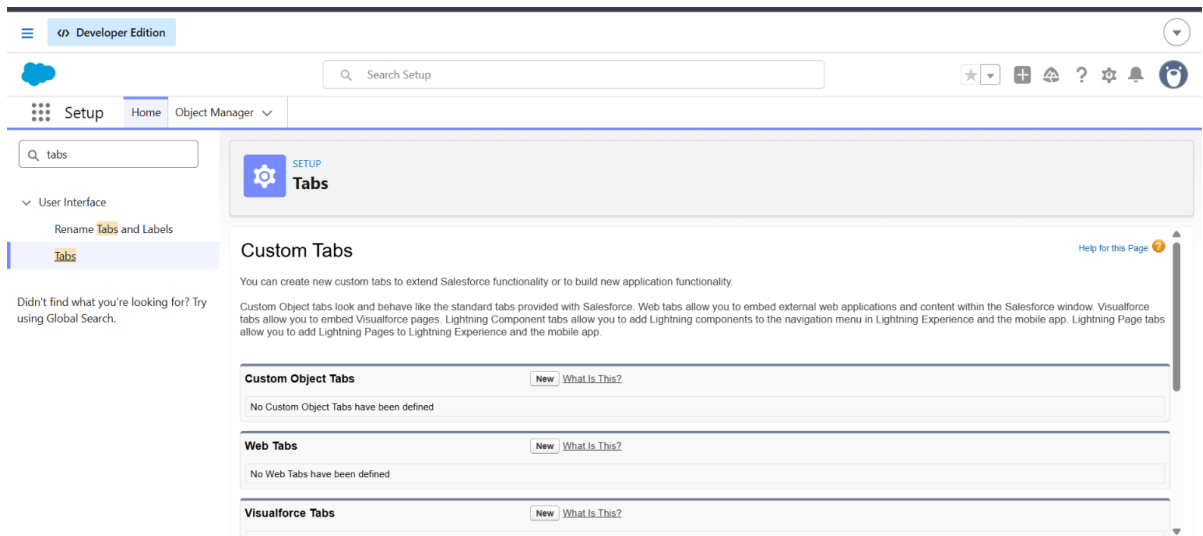
When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

 - ☒ Allow Sharing
 - ☒ Allow Bulk API Access
 - ☒ Allow Streaming API Access
- Deployment Status:**

[What is this?](#)

 - ☐ In Development
 - ☒ Deployed
- Search Status:**

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)



2. Fields

Custom fields were created for each object based on the specific information it needed to capture — for example, pricing and stock-related fields on Vehicle, contact fields on Dealer and Customer, and status/date fields on the transactional objects (Vehicle Order, Vehicle Test Drive, Vehicle Service Request).

Developer Edition

Setup

Home

Object Manager

Search Setup

Vehicle

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

10 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Dealer	Dealer__c	Lookup(Vehicle Dealer)	✓	
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Price	Price__c	Currency(18, 0)		
Status	Status__c	Picklist		
Stock Quantity	Stock_Quantity__c	Number(18, 0)		
Stock Quantity	Stock_Quantity__c	Number(18, 0)		
Vehicle Model	Vehicle_Model__c	Picklist		
Vehicle Name	Name	Text(80)	✓	

Developer Edition

Setup

Home

Object Manager

Search Setup

Vehicle Dealer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Dealer Code	Dealer_Code__c	Auto Number		
Dealer Email	Dealer_Email__c	Email		
Dealer Location	Dealer_Location__c	Text(18)		
Dealer Name	Name	Text(80)		✓
Dealer Phone	Dealer_Phone__c	Phone		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

Developer Edition

Setup

Home

Object Manager

Search Setup

Vehicle Service Request

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Created By	CreatedById	Lookup(User)		
Issue Description	Issue_Description__c	Text(18)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Service Date	Service_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)	✓	
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)	✓	
Vehicle Service Request Name	Name	Text(80)	✓	
Vehicle Service Request Status	Vehicle_Service_Request_Status__c	Picklist		

Developer Edition

Setup

Home

Object Manager

Vehicle Service Request

Details

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Created By	CreatedBy	Lookup(User)		
Issue Description	Issue_Description__c	Text(18)		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Service Date	Service_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)	✓	
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)	✓	
Vehicle Service Request Name	Name	Text(80)	✓	
Vehicle Service Request Status	Vehicle_Service_Request_Status__c	Picklist		

Developer Edition

Setup

Home

Object Manager

Vehicle Test Drive

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Test Drive Date	Test_Drive_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)	✓	
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)	✓	
Vehicle Test Drive Name	Name	Text(80)	✓	
Vehicle Test Drive Status	Vehicle_Test_Drive_Status__c	Picklist		

Developer Edition

Setup

Home

Object Manager

Vehicle Customer

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Preferred Vehicle Type	Preferred_Vehicle_Type__c	Picklist		
Vehicle Customer Address	Vehicle_Customer_Address__c	Text(18)		
Vehicle Customer Email	Vehicle_Customer_Email__c	Email		
Vehicle Customer Phone	Vehicle_Customer_Phone__c	Phone		
Vehicle Customer Name	Name	Text(80)	✓	

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Vehicle Order

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Order Date	Order_Date__c	Date		
Owner	OwnerId	Lookup(User,Group)		✓
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Order Name	Name	Text(80)		✓
Vehicle Order Status	Status__c	Picklist		

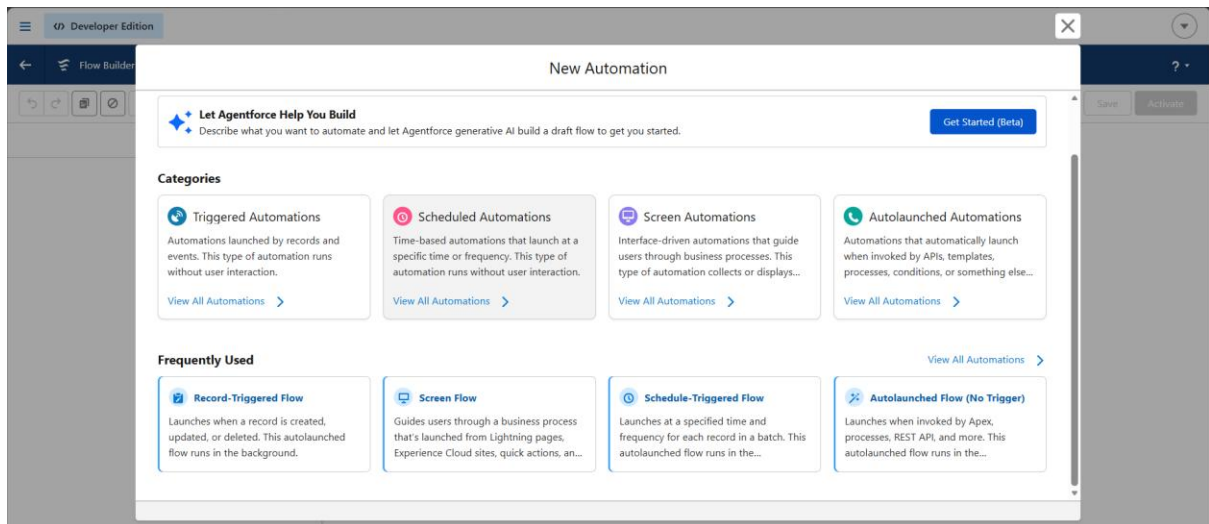
3. Validation Rule — Vehicle Stock

A validation rule was implemented to ensure an order cannot be confirmed when the associated vehicle does not have the required stock. This protects the ordering process from producing an invalid transaction — for example, attempting to confirm an order against a vehicle with zero stock quantity triggers the configured validation error rather than allowing the record to save.

4. Flow Automation

Salesforce Flow was used to automate selected business processes so that not every operation requires manual handling. The project includes two key flows:

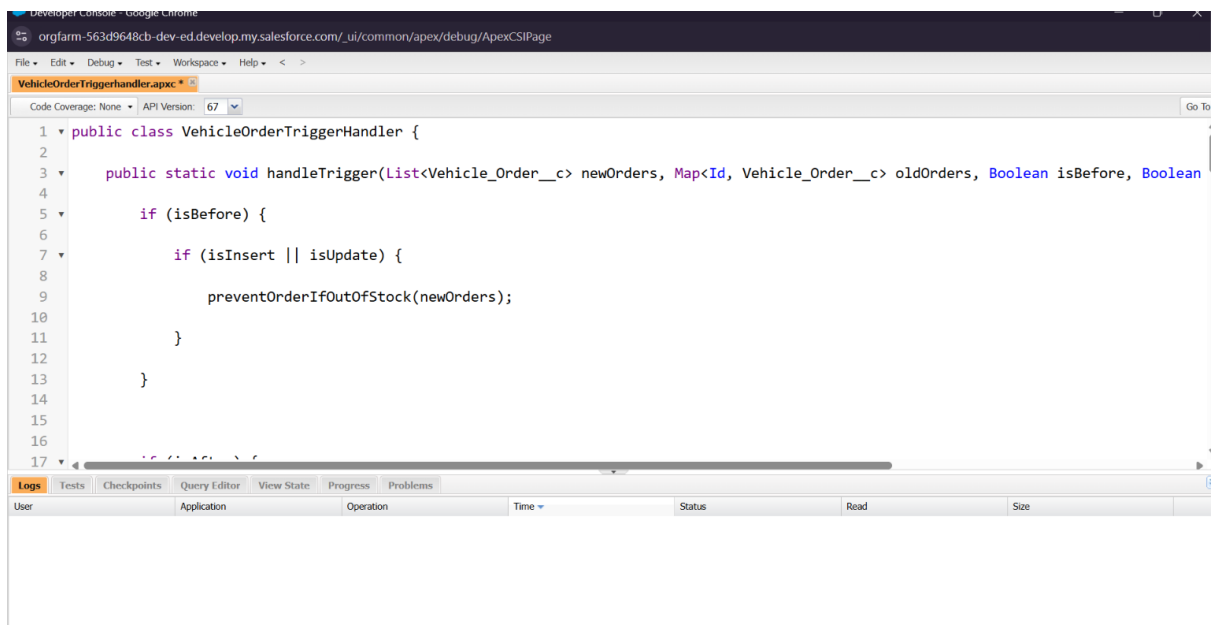
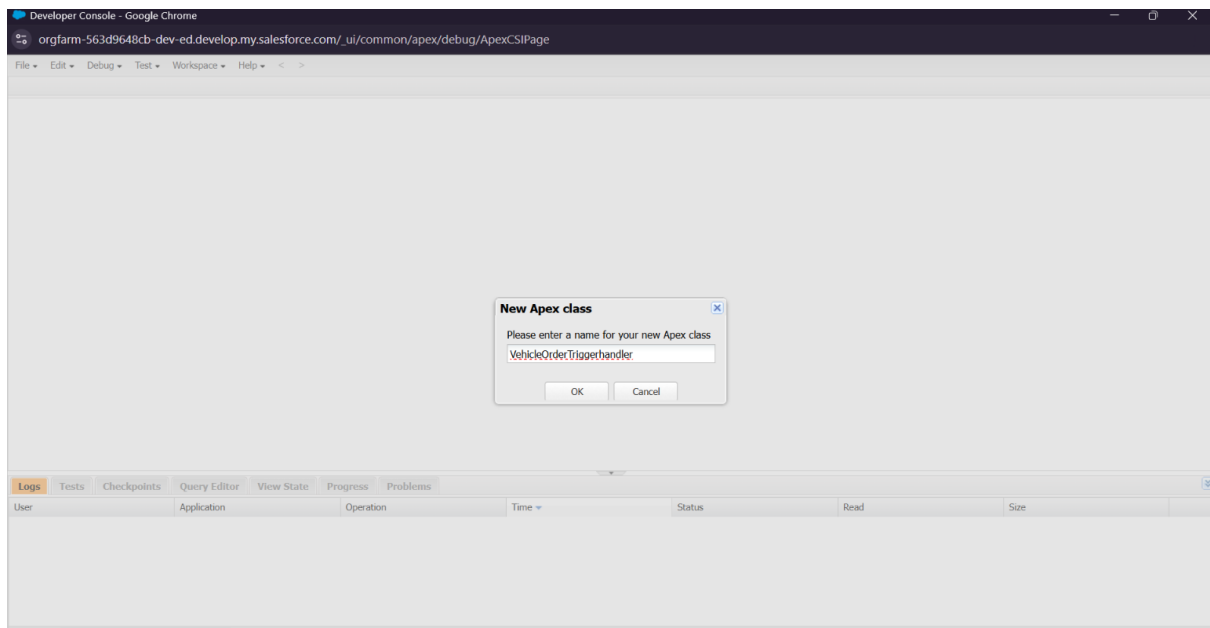
- Automatic dealer assignment — assigns the nearest or most appropriate dealer to a newly created order.
- Test-drive reminder — sends a reminder email to the customer ahead of a scheduled test drive.



Apex Classes and Triggers

1. Apex Classes

- Vehicle Order Processing — Apex classes handle the business logic associated with vehicle orders.
- Stock Validation — the classes check vehicle availability and enforce the defined stock conditions.
- Order Management — Apex logic supports processing and updating order records based on status.
- Reusable Business Logic — business logic is separated into dedicated Apex classes for easier maintenance and testing.
- Automation Support — Apex classes work alongside triggers and other Salesforce automation to complete required operations.

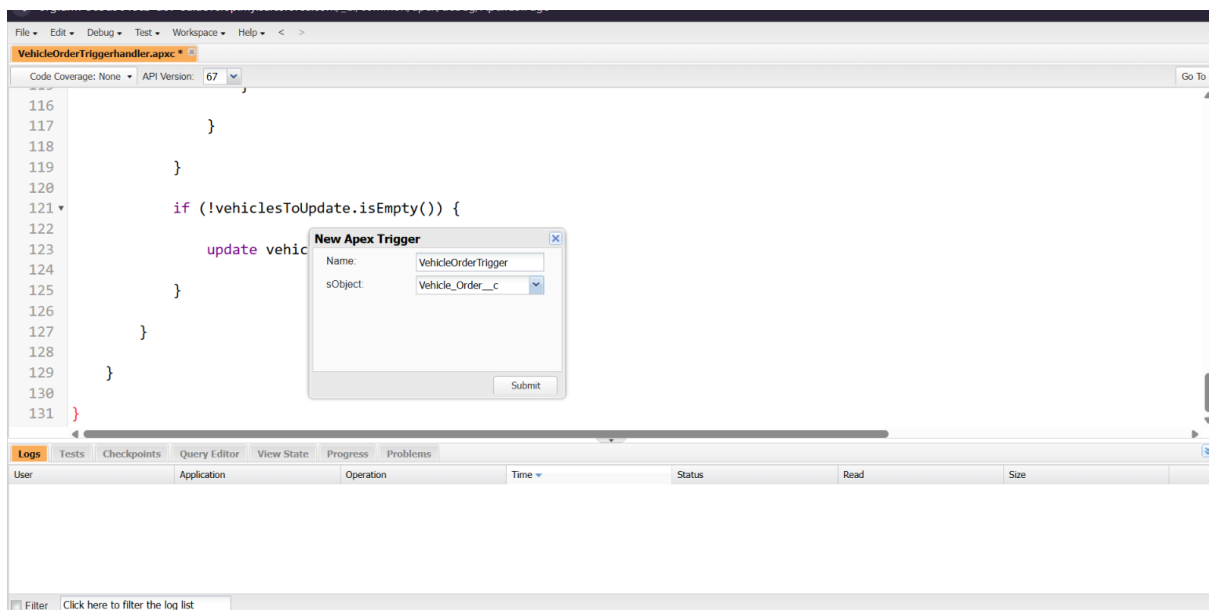


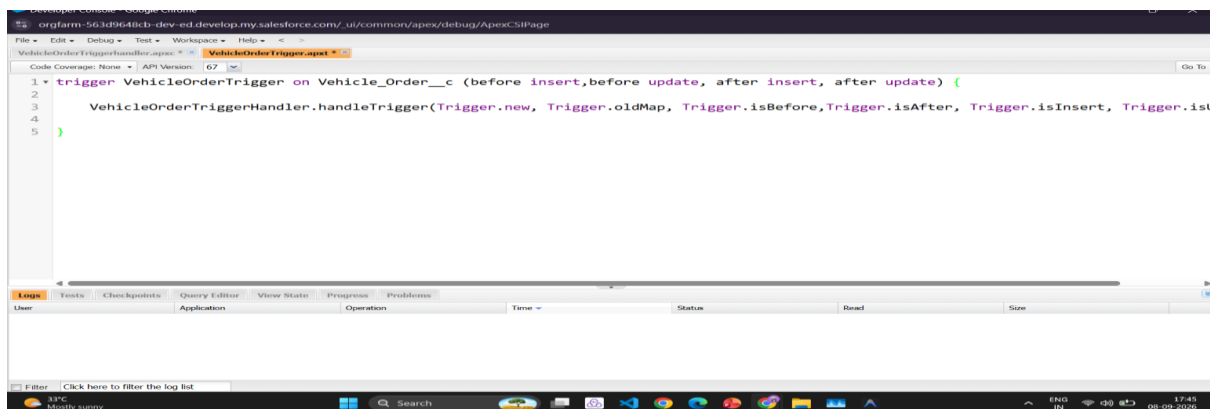
2. Apex Trigger

The vehicle order process is also supported by Apex-based backend logic. A trigger (VehicleOrderTrigger) responds to relevant order events and delegates its processing to a handler class (VehicleOrderTriggerHandler), keeping the trigger itself minimal. The handler is responsible for:

- Checking stock availability.
- Preventing invalid order processing.
- Updating vehicle stock after a confirmed transaction.
- Supporting order-status processing.

Organizing the logic this way — trigger plus handler — keeps the business rules separate from the trigger definition, making the code easier to maintain and extend.



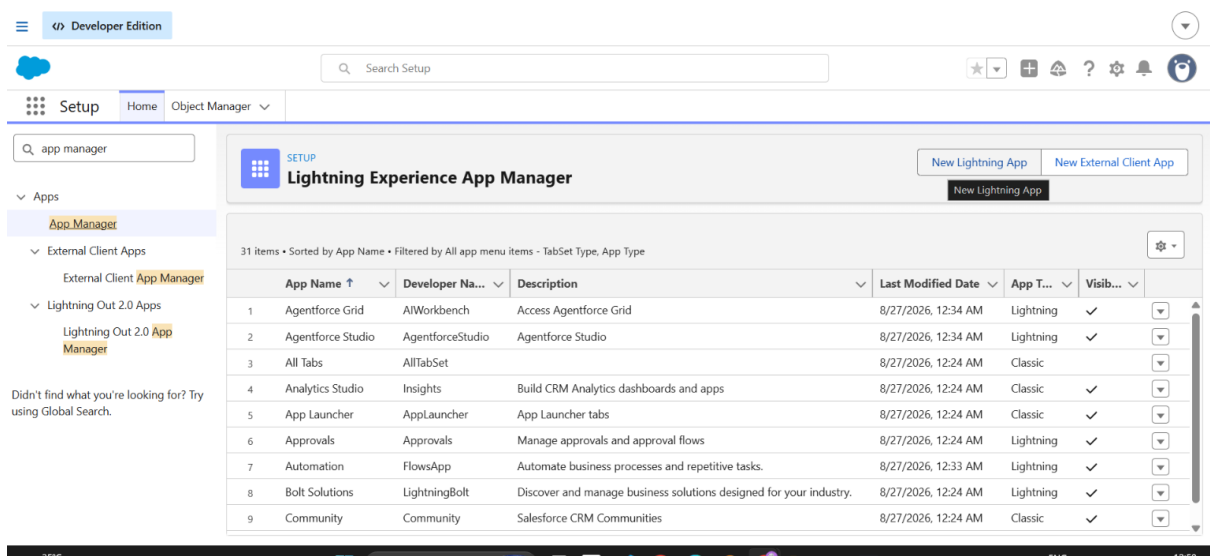


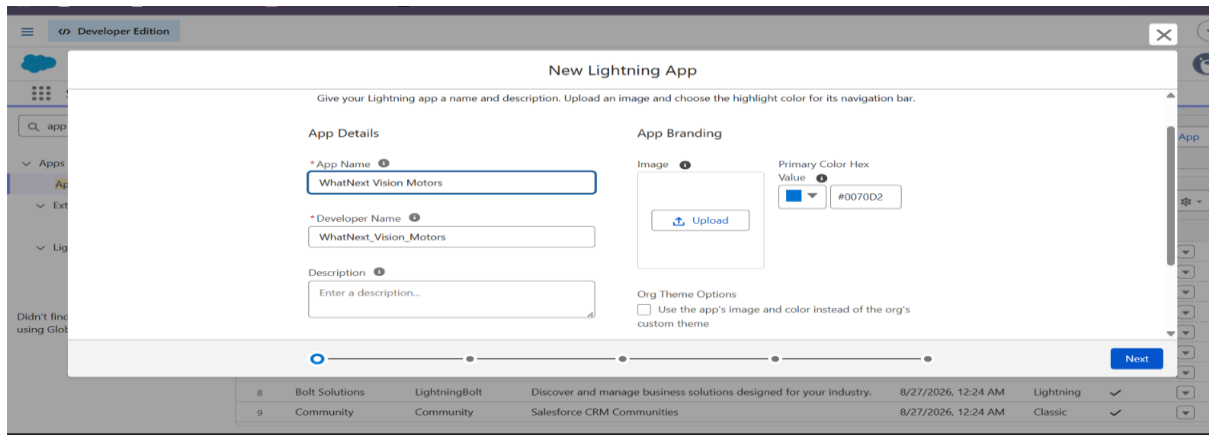
Phase 3: UI/UX Development & Customization

This phase focused on building a practical, user-friendly interface within Salesforce Lightning, covering the Lightning App, page layouts, Dynamic Forms, user management, reporting, and record pages.

1. Lightning App Setup through App Manager

A dedicated Lightning App was configured through App Manager to provide a centralized interface for managing Vehicles, Dealers, Customers, Orders, Test Drives, and Service Requests. This improves navigation and gives users straightforward access to the records and functions they need.



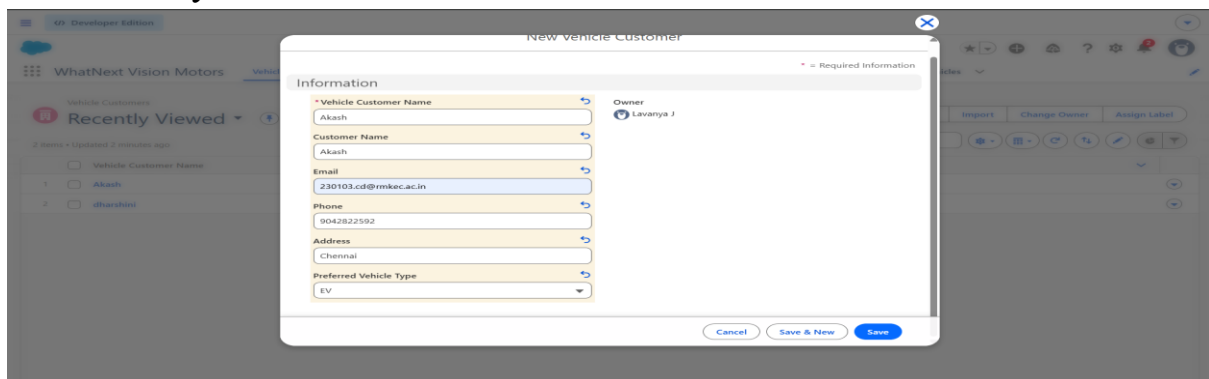


2. Page Layouts and Dynamic Forms

Page layouts were configured to arrange fields and related information so that users can see what's relevant without the record interface becoming cluttered. Dynamic Forms were applied wherever conditional field visibility was needed, so fields display based on the values or conditions already present on a record rather than all appearing at once.

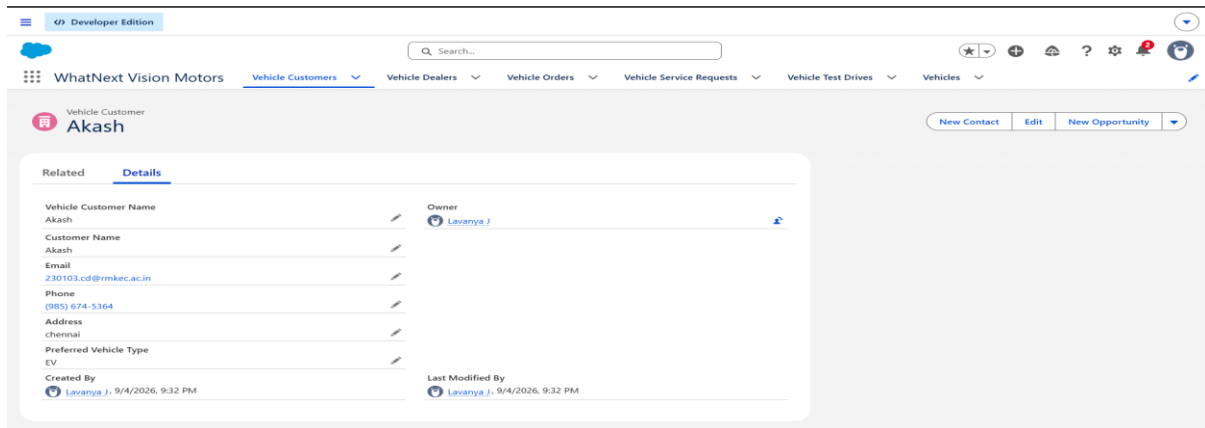
3. User Management

Users were configured in line with the access requirements of the application, ensuring each person can work with the records and functionality relevant to their role.



4. Reports and Dashboards

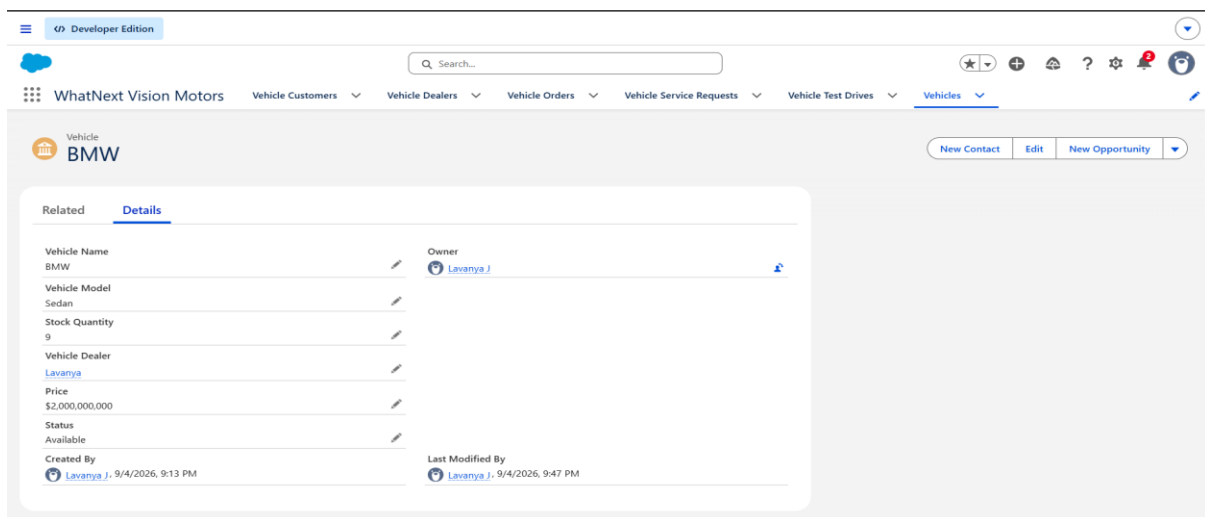
Reports were configured to provide structured, filterable views of Salesforce records. Dashboards present selected report data visually, helping users get a quick read on business activity without having to review individual records one at a time.



5. Lightning Record Pages

Lightning Record Pages were configured to give users a useful view of each individual record. Key components include:

- Record details
- Highlights panel
- Related lists



Phase 4: Data Migration, Testing & Security

The fourth phase covered populating the CRM with data, verifying the implemented functionality, and putting the appropriate security controls in place.

1. Data Loading

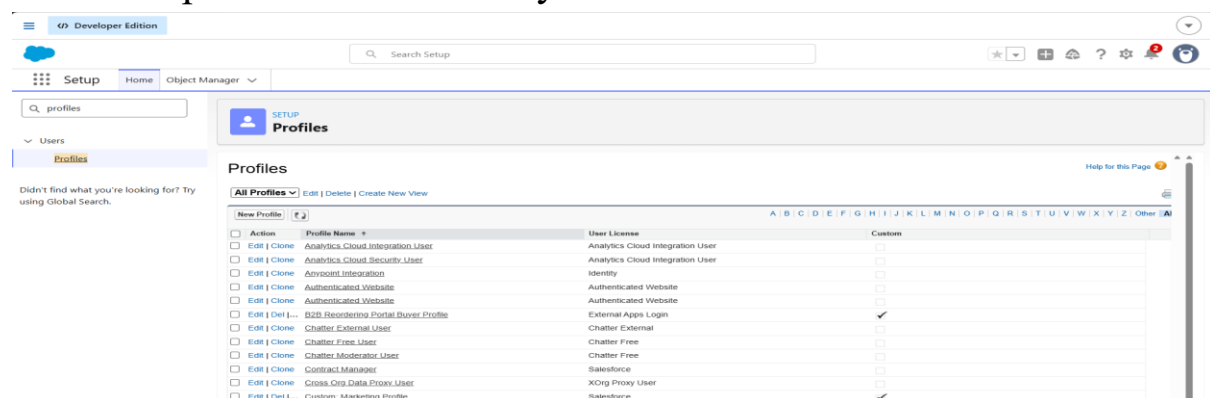
Initial data was prepared in a structured format before being loaded into Salesforce. The process followed these steps:

1. Prepare sample records.
2. Map the data fields to the corresponding Salesforce fields.
3. Load the required records.
4. Verify that records were created correctly.
5. Confirm relationships between associated records.

Depending on the object and volume of records involved, either the Salesforce Data Import Wizard or Data Loader was used.

2. Profiles

Profiles establish the baseline permission structure for Salesforce users, determining the level of access they have to applications, objects, fields, and other platform functionality.



3. Roles and Role Hierarchy

The role hierarchy reflects the organizational structure used for record-access decisions. A representative structure for this project is:

CEO → Sales Manager → Sales Representative

The role hierarchy actually reflected in the Salesforce org corresponds to the reporting structure used for this implementation.

The screenshot shows the Salesforce Setup interface for the 'Users' section. The left sidebar contains navigation links for Analytics Groups, Permission Set Groups, Permission Sets, Profiles, Public Groups, Queues, Roles, and User Management Settings. The main content area displays the 'User Detail' for a user named Laviya J. The user's profile is 'Salesforce System Administrator', and their role is 'Salesforce System Administrator'. The user's email is '230103.cd@rmkcc.ac.in' and their username is '230103.cd.76068596989@agentforce.com'. The user's company is 'RMK Engineering College' and their department is 'Department'. The user's address is 'Division', 'Address', 'Time Zone (GMT-07:00) Pacific Daylight Time (America/Los_Angeles)', 'Locale English (United States)', 'Language English', and 'Delegated Approver Manager'. The user's role is 'Salesforce System Administrator' and their profile is 'Salesforce System Administrator'. The user's status is 'Active' and their last login was '2023-07-07 10:00:00'. The user's last login IP address is '172.16.17.1' and their last login location is 'San Francisco, CA, United States'.

4. Permission Sets

Permission sets were used to grant additional access to users who require specific Salesforce functionality beyond what their base profile provides, without having to modify the profile itself.

The screenshot shows the Salesforce Setup interface for the 'Permission Sets' section. The left sidebar contains navigation links for Users, Permission Set Groups, Permission Sets, Custom Code, and Custom Permissions. The main content area displays the 'Permission Sets' list. The list includes columns for Action, Permission Set Name, Description, and License. The list contains 10 permission sets, all of which are 'Clone' actions. The permission sets are: 'Access Agentforce Default Agent', 'Agentforce Default Agent', 'Agentforce Default Agent Admin', 'Agentforce Service Agent Configuration', 'Agentforce Service Agent Object Access', 'Agentforce Service Agent Secure Base', 'Agentforce Service Agent User', 'Agentforce Near Core CRM', 'Agentforce Near Core Marketing', and 'Agentforce Near Core Personalization'. The license for all permission sets is 'Agentforce (Default)'.

5. Sharing Rules

Sharing configuration determines how records are accessed across different users. The sharing model was designed to align with each user's responsibilities, so staff can work with the information their role requires while restricted records remain appropriately protected.

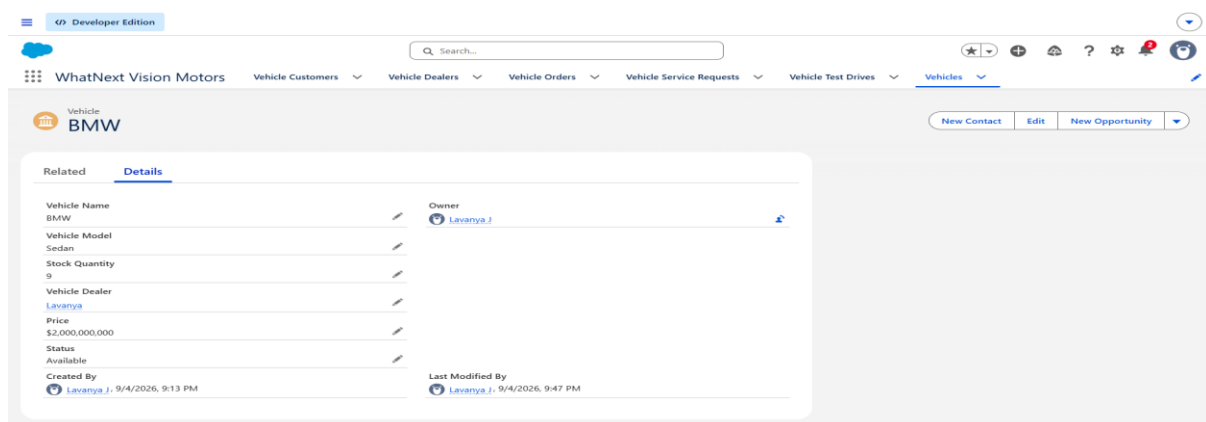
Testing

Test cases were prepared for the application's key Salesforce features — including record creation, validation rules, and automated flows — each backed by input and output screenshots.

Test Case 1 — Vehicle Creation

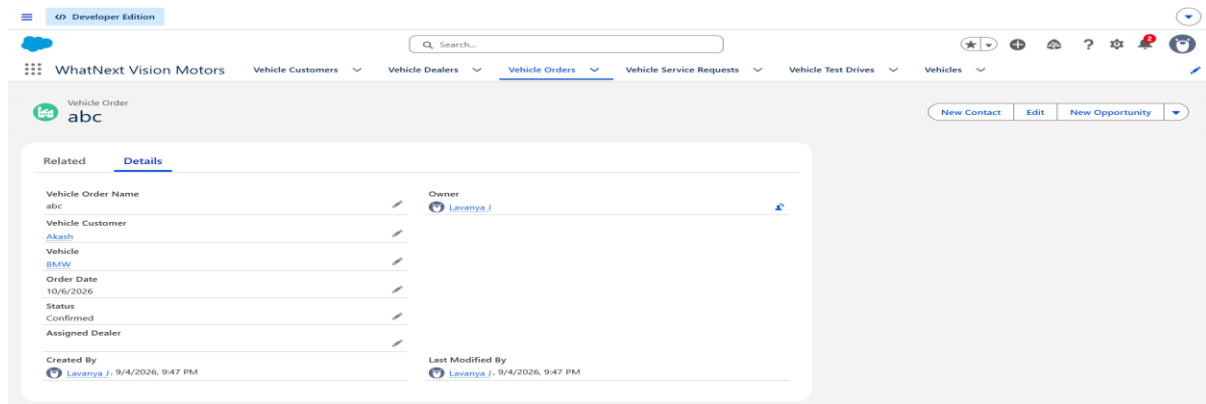
Objective: verify that a valid vehicle record can be created.

- Vehicle Name: BMW
- Model: Sedan
- Stock Quantity: 9
- Price: \$2,000,000,000
- Status: Available
- Dealer: Lavanya



Test Case 2 — Out-of-Stock Order

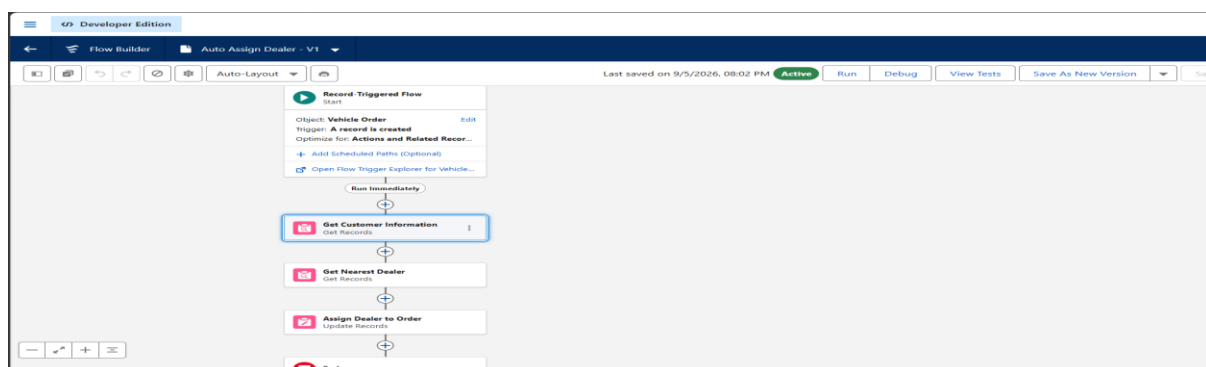
Objective: verify that an order cannot be confirmed when the selected vehicle has no available stock.

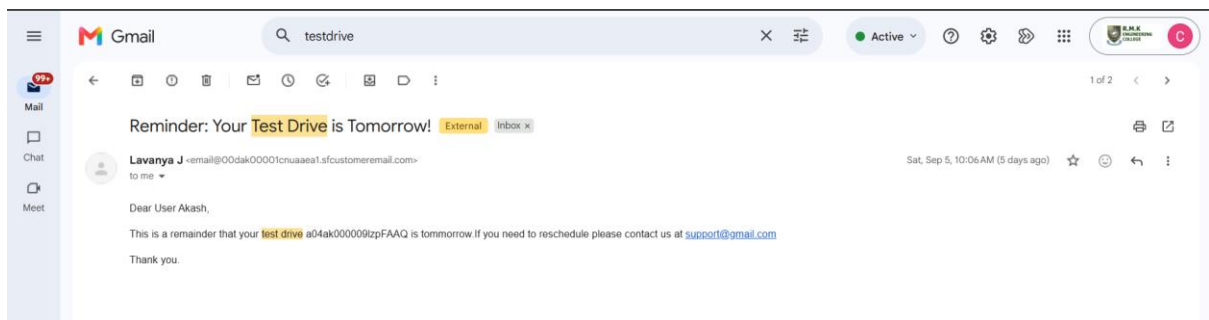
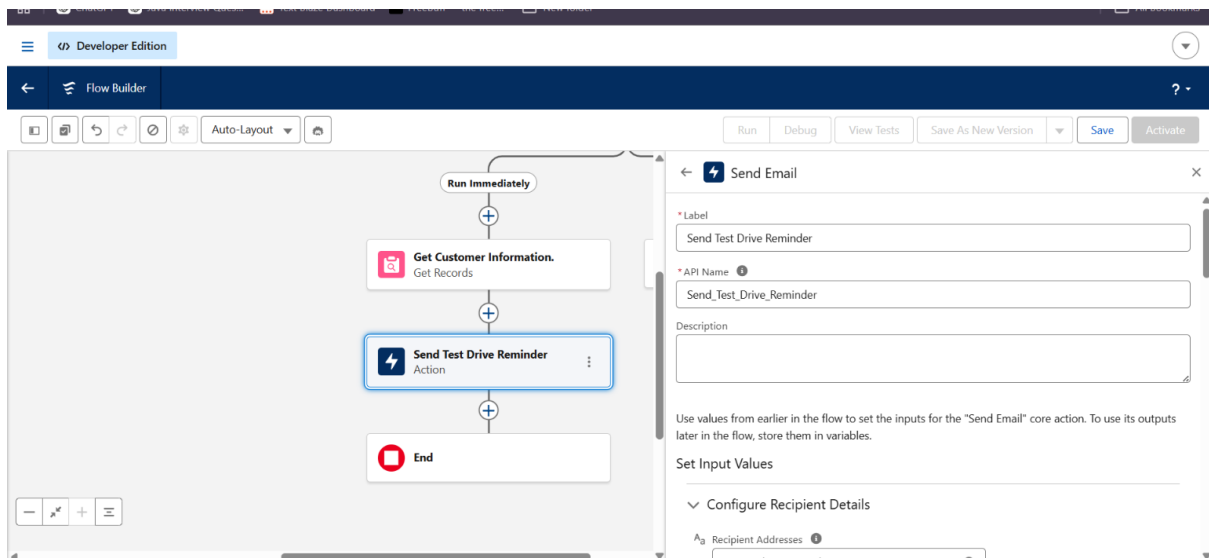


Test Case 3 — Test Drive Reminder Email

Objective: verify the automated reminder process for an upcoming test drive.

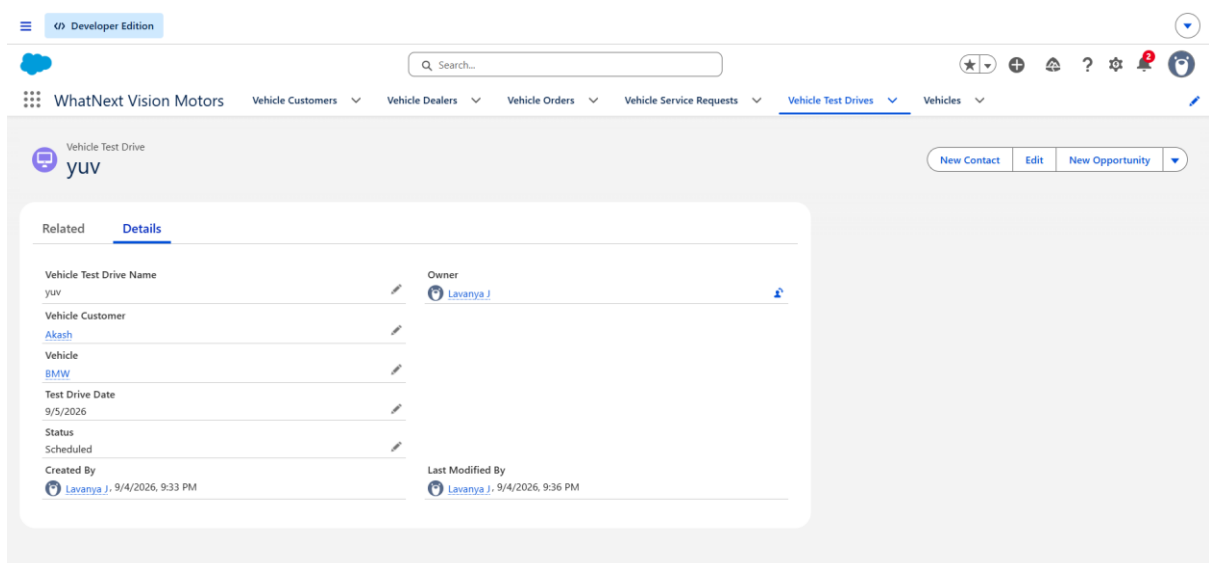
- Customer: existing customer
- Customer Email: valid email address
- Status: Scheduled
- Test Drive Date: following day





Test Case 4 — Pending Order Batch Processing

Objective: verify that a pending order can be processed once the associated vehicle's stock becomes available.



Phase 5: Deployment, Documentation & Maintenance

The final phase addressed moving Salesforce components to their target environment, monitoring the live application, handling issues as they arise, and maintaining the system over time.

1. Deployment Strategy

Salesforce metadata can be promoted between environments using supported deployment mechanisms such as change sets or Metadata API-based tooling. For this project, the components involved in a deployment include:

- Custom objects
- Custom fields
- Object relationships
- Validation rules
- Flows
- Apex classes
- Apex triggers
- Lightning pages
- Reports

2. Maintenance and Monitoring

Several built-in Salesforce tools support the ongoing health of the application:

- Apex Jobs — used to monitor batch and scheduled Apex execution.
- Debug Logs — used to investigate unexpected behaviour in Apex and other Salesforce operations.
- Flow Errors — reviewed to identify problems within automated processes.

- User Access Reviews — carried out periodically to confirm users retain only the access their role requires.

3. Troubleshooting Approach

A structured method is followed whenever an issue is identified:

1. Identify the Problem — determine which Salesforce feature is producing the unexpected result.
2. Reproduce the Issue — repeat the operation using controlled test data.
3. Review Configuration — inspect the relevant object, fields, validation rules, Flow, permissions, or Apex logic.
4. Check Logs — review debugging information where automation or Apex is involved.
5. Correct the Issue — modify the configuration or code responsible for the problem.
6. Retest — run the original scenario again to confirm the expected result.
7. Deploy the Correction — move the corrected configuration to the required environment following the chosen deployment process.

Future Enhancements

Looking beyond the current implementation, the application could be extended with the following capabilities:

- Chatbot integration to handle routine customer queries automatically.
- AI-based vehicle recommendations tailored to individual customer preferences.
- Deeper analytics and forecasting to support dealer-level demand planning.

Conclusion

The WhatNext Vision Motors project delivered a Salesforce CRM environment capable of organizing a range of automotive business activities within a single platform. It demonstrates how Salesforce can be tailored to represent business-specific entities and processes from end to end, combining declarative configuration with custom Apex development.

The major outcomes of the project include:

- Automotive records organized through purpose-built Salesforce objects.
- Relationships connecting customers, vehicles, dealers, orders, and related activities.
- Vehicle stock conditions checked automatically during order processing.
- Dealer assignment handled through automation rather than manual lookup.
- Test-drive reminders generated without requiring staff intervention.
- Pending orders processed automatically once the required stock becomes available.
- Lightning pages delivering a structured, organized user experience.
- Layered security controls providing appropriately restricted access.
- Reports and dashboards supporting ongoing business monitoring.
- Documented test cases confirming that the implemented functionality behaves as expected.

Overall, the project provided practical, hands-on experience in combining Salesforce's declarative tools with Apex development to solve a real-world business problem.